

PYTHON PROGRAMMING LAB MANUAL

Week 3 — Control Structures (Conditional Statements and Loops)

Program	Mechanical Engineering	Semester	IV
Course Name	Programming Concepts in Mechanical Engineering	Course Code	25ME34I
Week	3	Duration	3 Hours (Practice Session)
Course Outcome	CO-1: Use appropriate syntaxes while writing program.	Program Outcomes	PO-1, PO-2, PO-4

Learning Objectives

- Write decision-making programs using if, if-else, elif, and nested if statements.
- Use for loops and while loops to repeat a set of instructions.
- Apply loop control statements — break, continue, and pass — to control loop execution.
- Combine conditionals and loops to solve small engineering problems.

Quick Theory Recap

- if / elif / else: run a block of code only when a condition is True; elif checks additional conditions in order.
- Nested if: an if statement placed inside another if (or else) block, used when a decision depends on more than one condition, checked in stages.
- for loop: repeats a block a fixed number of times, typically over range() or a sequence.
- while loop: repeats a block as long as a condition remains True — used when the number of repetitions is not known in advance.
- break: exits the loop immediately. continue: skips the rest of the current iteration and moves to the next. pass: a placeholder that does nothing (used to keep syntactically valid empty blocks).

Practical Exercises

Exercise 1: Even or Odd Number

Aim: Write a Python program to check whether a number entered by the user is even or odd.

Program:

```
# Program to check whether a number is even or odd

# Input from user
num = int(input("Enter a number: "))

# Decision making
if num % 2 == 0:
    print("The number is Even")
```

```
else:
    print("The number is Odd")
```

Sample Output

```
Enter a number: 17
The number is Odd
```

Explanation: % is the modulus operator — it gives the remainder of division. If the remainder is 0, the number is Even; otherwise it is Odd.

Exercise 2: Multiplication Table Using a for Loop

Aim: Print the multiplication table of a number entered by the user using a for loop.

Program:

```
# Program to print multiplication table of a given number

# Input from user
num = int(input("Enter a number: "))

# Using for loop
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

Sample Output

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

*Explanation: range(1, 11) runs the loop from 1 to 10; num * i calculates the multiplication; the for loop repeats the print statement 10 times.*

Exercise 3: Factorial of a Number Using a while Loop

Aim: Calculate the factorial of a number entered by the user using a while loop.

Concept:

- Factorial of n ($n!$) = $1 \times 2 \times 3 \times \dots \times n$

Program:

```
# Program to calculate factorial of a number using while loop

# Input from user
num = int(input("Enter a number: "))
fact = 1
i = 1

# While loop to calculate factorial
while i <= num:
    fact = fact * i
    i = i + 1

# Display result
print("Factorial of", num, "is:", fact)
```

Sample Output

```
Enter a number: 5
Factorial of 5 is: 120
```

Explanation: The while loop repeats until the condition $i \leq \text{num}$ becomes false, multiplying fact by i on every pass. The variable fact stores the running result.

Exercise 4: Nested if — Grading System

Note: Added to this manual: nested if statements are listed in the Week 3 syllabus but were not covered in the original draft.

Aim: Use a nested if statement to assign a grade based on marks scored in a test.

Program:

```
# Program to assign a grade using nested if statements

marks = int(input("Enter marks (out of 100): "))

if marks >= 35:
    if marks >= 90:
        grade = "A"
    else:
        if marks >= 75:
            grade = "B"
        else:
            grade = "C"
else:
    grade = "Fail"

print("Grade:", grade)
```

Sample Output

Enter marks (out of 100): 82
Grade: B

Explanation: The outer if first checks whether the student has passed (marks >= 35). Only if that is True does an inner (nested) if decide which grade band the marks fall into.

Exercise 5: Loop Control Statements — break, continue, pass

Note: Added to this manual: break, continue, and pass are listed in the Week 3 syllabus but were not covered in the original draft.

Aim: Demonstrate break, continue, and pass while scanning a list of machine sensor readings.

Program:

```
# Program to demonstrate break, continue, and pass

readings = [45, 50, -1, 60, 999, 55]

for value in readings:
    if value == -1:
        # faulty sensor reading -> skip it and continue
        continue
    elif value == 999:
        # critical fault -> stop scanning immediately
        print("Critical fault detected. Stopping scan.")
        break
    elif value > 100:
        pass # placeholder for future high-value handling
    else:
        print("Reading OK:", value)
```

Sample Output

```
Reading OK: 45
Reading OK: 50
Reading OK: 60
Critical fault detected. Stopping scan.
```

Explanation: continue skips the faulty (-1) reading; break stops the loop the moment a critical value (999) is found, so the reading after it (55) is never checked; pass is a placeholder that does nothing.

Additional Practice Programs

The following short programs were part of the original Week 3 draft. They are retained here, cleaned up and given sample outputs, as extra if-else and loop practice.

Part A: if-else — Practice Set

A.1 Positive / Negative / Zero

```
n = int(input())

if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

Sample Output

```
5
Positive
```

Explanation: > checks positive, < checks negative, and 0 is handled separately by else.

A.2 Largest of Two Numbers

```
a = int(input())
b = int(input())

if a > b:
    print(a)
else:
    print(b)
```

Sample Output

```
15
9
15
```

Explanation: Compares the two values using > and prints the larger one.

A.3 Voting Eligibility

```
age = int(input())

if age >= 18:
    print("Eligible")
else:
    print("Not Eligible")
```

Sample Output

```
20
Eligible
```

Explanation: The minimum voting age is 18.

Part B: Loops — Practice Set

B.1 Print 1 to 10

```
for i in range(1, 11):  
    print(i)
```

Sample Output

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

Explanation: range(1, 11) generates numbers 1 to 10, printed one per line.

B.2 Even Numbers from 1 to 50

```
for i in range(2, 51, 2):  
    print(i)
```

Sample Output

```
2  
4  
6  
8  
10  
...  
48  
50
```

Explanation: The third value in range(2, 51, 2) is the step size — stepping by 2 from 2 to 50 prints only even numbers.

B.3 Reverse a Number

```
n = int(input())  
rev = 0  
  
while n > 0:
```

```

    rev = rev * 10 + n % 10
    n //= 10

print(rev)

```

Sample Output

```

1234
4321

```

Explanation: $n \% 10$ extracts the last digit each time, which is appended to rev ; $n //= 10$ removes that digit from n .

B.4 Count Digits

```

n = int(input())
count = 0

while n > 0:
    count += 1
    n //= 10

print(count)

```

Sample Output

```

4321
4

```

Explanation: One digit is removed from n on every pass of the loop; $count$ keeps track of how many passes were needed.

Self-Practice (To Be Completed Individually)

Attempt the following on your own and get your output verified by the faculty:

- Write a Python program that uses if-elif-else to check whether a machine component's temperature is Normal, Warning, or Critical.
- Write a Python program using a for loop to print the sum of the first n natural numbers entered by the user.
- Write a Python program that uses a nested if to check whether a triangle (given three sides) is equilateral, isosceles, or scalene.
- Write a Python program that scans a list of numbers and uses continue to skip negative values while printing the rest.
- Write a Python program that uses break to stop a while loop as soon as a sensor reading exceeds a safe limit.

Reference: Week 3 syllabus content (25ME34I — Programming Concepts in Mechanical Engineering, DTE Karnataka).